Parallel circuits

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate branch currents to a shared potential difference.

SUCCESS CRITERIA - I can...

1. Identify branches connected across the same two nodes.
2. Use equal voltage across parallel branches.
3. Add branch currents to find supply current.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “parallel circuit” and “node” to explain your answer. Sentence starters are optional.

parallel circuit: branches connected across the same two nodes

node: a shared electrical connection point

current: rate of charge flow

01 Warm-up: bring back an earlier idea

A current of **3 A** flows for **5 s**. **Find the charge** that passes.

Word help: charge = a property responsible for electric interactions

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\mathbf{1}/R_{total} = \mathbf{1}/R_1 + \mathbf{1}/R_2$; $I_{total} = I_1 + I_2$; $V_1 = V_2$. Parallel branches connect to the same two nodes and share voltage.

Worked reasoning: Two parallel branches of **6** and **3 ohms** each have **6 V** across them. Branch currents are **1** and **2 A**; source current is **3 A**.

Your turn: Draw the two branches and **mark where the currents split** and recombine.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Mark the two shared nodes in the parallel diagram. If one branch opens, can the other remain powered? Explain.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .



Circuit symbols: long/short lines = battery; rectangles = resistors.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A circuit board has **3 Ω** and **6 Ω** resistors connected in parallel across an ideal **6 V** battery. **Label the voltage across each** branch, calculate both currents and **show how they combine** at the source.

Word help: ideal = using stated simplifying conditions

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

05 Explain your reasoning

Why can the equivalent resistance be smaller than either branch resistance?

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A bundle of parallel rays passes through a lens and becomes **4** times as wide farther beyond it, with rays spreading immediately after the lens. **Name the lens type** and **sketch the rays**.
Draw the rays used in your explanation.

Word help: lens = a transparent element that changes light-ray directions

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: ① ② ③

07 Exit check

Two parallel branches draw **0.2 A** and **0.3 A**.
1. **Find the current supplied** to their junction.
2. **Label the circuit** and **show the current path or** paths.

Sentence starter: I found … [unit]. This value means … in this situation.



Label component values, current and voltage as needed.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. **Add a third 6 ohm** branch to this ideal supply.
2. **Predict each existing branch current** and the new total current.
3. **Label the circuit** and **show the current path or** paths.

Word help: ideal = using stated simplifying conditions

Sentence starter: I predict … . I used … to make this prediction because … .

Draw the complete circuit and label shared nodes.

Marzano check - circle: ① ② ③